

Jackson Opoku

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PROFESSIONAL EXPERIENCE

Senior Photolithography Photomask Process Engineer

Taiwan Semiconductor Manufacturing Company (TSMC), Arizona (FAB21)

June 2022 – Present | Phoenix, Arizona

- Lead and supervise technicians across high-volume manufacturing, training them on selected engineer-level tasks and coordinating with cross-functional, cross-cultural teams.
- Supported production ramp-up to **25,000 wafers per month** while reducing repeating wafer defects by **20%** through focused defect defense actions.
- Improved throughput from photo to etch by lowering rework and wafer scrap at the photo stage.

Photolithography Photomask Technician Supervisor

Taiwan Semiconductor Manufacturing Company (TSMC), Arizona (FAB21)

June 2025 – January 2026 | Phoenix, Arizona

- Supervised a technician team in a leading-edge fab environment, supporting production ramp-up to **25,000 wafers per month** during early fab operations.
- Developed a **technician upskilling program** that trained technicians to safely perform selected engineer-owned tasks, driving a **first-in-TSMC-history initiative** that fully staffed off-shifts without an engineer on duty.
- Strengthened technician autonomy and cross-functional collaboration, improving operational continuity, response time, and tool availability.

Sponsor for Holon LEXa-20 EDX – EUV Tool Defect Analysis

Taiwan Semiconductor Manufacturing Company

January 2023 – Present

- Sponsored the Holon LEXa-20 EDX tool to identify elemental composition of reticle fall-on defects, leading investigations that pinpointed scanner component degradation for targeted corrective action.
- Improved EUV scanner throughput and equipment efficiency by resolving fall-on defects and deploying a Pump–Vent Containment Strategy that cut tool downtime from particle events.

Mask Frontside Defect Inspection Tools (Lasertec X812/X8U and KLA SL670e)

Taiwan Semiconductor Manufacturing Company

January 2023 – March 2024 | Phoenix, AZ

- Spearheaded the release of Lasertec Optical/EUV and KLA inspection tools into production across Taiwan and Arizona fabs.
- Engineered automated SEM imaging workflows that sharpened detection sensitivity and cut false-defect counts, and authored SOPs that reduced frontside defect rates on newly taped-out masks from EBO.

EUV Mask Backside Defect Inspection Tool (Lasertec BASIC BX100)

Taiwan Semiconductor Manufacturing Company

January 2023 – March 2024 | Phoenix, AZ

- Boosted fab-wide EUV reticle backside inspection performance, cutting RBI reject alarms and lifting productivity by **50%**.
- Built and delivered BX100 training and SOPs fab-wide to prevent scanner cross-contamination from backside defects.

Sponsor for Wafer Image Acceptance for Risky Reticle Defects (WIA)

Taiwan Semiconductor Manufacturing Company

January 2023 – Present | Phoenix, AZ

- Led the WIA process, validating reticle fall-on defects via inline exposure conditions and detailed image analysis, and served as primary liaison between the reticle shop and production teams on reticle standards.
- Oversaw mask-to-wafer image evaluations and quality compliance while driving continuous improvement that enhanced yield and process robustness.
- Cut WIA cycle time from the company standard of **8 hours to 6 hours** by enabling technicians to handle basic engineer-focused WIA actions.

Sponsor for Particle per Reticle Pass (PRP) Tracking – EUV Lithography

Taiwan Semiconductor Manufacturing Company

January 2023 – Present

- Sponsored and led PRP tracking, a critical metric monitoring EUV reticle fall-ons per 10,000 wafer moves, collaborating with ASML defectivity teams and EUV equipment engineers on tool selection.
- Increased EUV availability by approximately **10%** through proactive degradation detection, and developed PRP-based tool prioritization strategies for urgent customer products.

- Improved lithography stability and strengthened cross-functional decision-making by leveraging PRP data for short- and long-term production goals.

Sponsor for Gudeng Barcode Reader Tool – NTO Reticles & Pellicle Rupture Detection

Taiwan Semiconductor Manufacturing Company

May 2024 – Present

- Delivered production release in under two months by integrating dual algorithms that significantly reduced false NG alarms.
- Improved pellicle rupture detection accuracy and increased EUV scanner uptime through faster rupture confirmation and recovery.

ACADEMIC EXPERIENCE

Graduate Research Assistant – Manufacturing Innovation Centre

Arizona State University

August 2021 – May 2022

- Conducted mechanical testing of small-scale materials (e.g., ECAP copper) on the MTS Tytron 250 and analyzed stress–strain data to determine usability and optimal performance.

Graduate Teaching Assistant – Senior Capstone Design

Arizona State University

August 2021 – April 2022

- Supervised 29 senior capstone teams through structured product development, including funded projects sponsored by Northrop Grumman.
- Graded assessments and delivered timely feedback to support effective communication and project execution.

EDUCATION

Master of Science in Mechanical Engineering

Arizona State University, Tempe, Arizona

GPA: 4.0 | Mastercard Foundation Scholar

International Accelerated Degree Program (3+1+1)

Dean's List: Fall 2020, Spring 2021

Bachelor of Science in Mechanical Engineering

Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Completed final year at Arizona State University (ABET-accredited)

First Class Honors | Excellent Student Award (Spring 2019, 2020)

TECHNICAL SKILLS & TRAINING

- EUV Lithography, Reticle Defectivity, PRP & WIA
- Experienced in Optical and EUV Photomask defect defense for N3/N4/N5 process nodes
- Holon Lexa-20, Lasertec BX100, Lasertec X8U and Lasertec X812, Hitachi CDSEM
- SQL, MATLAB, Mathematica
- Lean Six Sigma Certified

CONFERENCES & PROFESSIONAL DEVELOPMENT

SPIE Photomask Technology + EUV Lithography Conference

September 2026

- Nominated by TSMC's EUV Lithography department to serve as the **seed representative for the Photomask section**.